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Curriculum Vitae

Atharv Maheshwari

Third-Year Undergraduate at IIT Bombay

Homepage
LinkedIn

Research Interests

I have a keen interest in Condensed Matter Physics and Quantum Technology. I want to explore the phenomena that emerge from various quantum materials and devices and their applications in modern-day technologies.

Education

- **B.Tech, Engineering Physics**, Indian Institute of Technology Bombay **2023 – 27**
  GPA: **9.57/10.00**, with **Honors**
  Pursuing a **Minor** in Electrical Engineering

Research Experience

Ion Traps and its Mechanical Analog **Jan 2025 - Jul 2025**
Prof. Kantimay Das Gupta | Department of Physics | IIT Bombay Project Documents

- Studied systems that trap ions in oscillating electric fields (**Paul traps**), creating a 3D pseudo-potential well using electrodes of specific geometries, and applied concepts from Floquet theory and numerical analysis
- Designed and fabricated a **3D-printed rotating-saddle** mechanical analog of ion trap, using Fusion360 software
- Analytically determined and experimentally verified the angular velocity for stability of sphere on rotating saddle
- Experimentally achieved **bound trajectory** for a sphere lasting nearly **280 seconds** at optimal angular velocity
- Derived the **effective trajectory** of **centre of mass** for an extended object and refined surface geometry to make the center of mass move along the desired hyperbolic-paraboloid (saddle) trajectory

Role of Nonlinear Resonances in Protection of Qubits **Aug 2024 - Present**
Prof. Sudhir Ranjan Jain | UM-DAE Centre for Excellence in Basic Sciences Project Documents

- Analyzing stable **elliptical** and unstable **hyperbolic fixed points** in macroscopic quantum systems, such as **Josephson junctions**, to understand system dynamics under varying parameters and optimize its stability
- Studying effects of perturbations in the system based on fundamentals of **Kolmogorov-Arnold-Moser** theorem
- Optimizing parameters to inhibit **information leakage** and suppress **decoherence** caused by quantum tunneling

Technical Experience

Mars Rover Team, IIT Bombay | *Guide: Prof. P.J. Guruprasad* **Oct 2023 - Present**

- Part of a 40-member team developing a semi-autonomous rover for traversing difficult terrains
- Ranked **1st** among Indian teams at **University Rover Challenge 2023**, USA, against 100+ international teams
- Ranked **2nd in European Rover Challenge (ERC)** 2025, contested by over 24 teams across the world
- Secured **1st position** in **Astrobiology Expedition Mission** at International Rover Challenge, 2023, conducted by Space Robotics Society at BITS Pilani, Goa

Senior Design Engineer, Robotic Arm Subsystem **Oct 2023 - Apr 2025**

- Designed a **5 DOF** highly maneuverable robotic arm for doing various tasks efficiently and precisely using a custom designed zero backlash cycloidal gearbox, replacing the redundant 6 DOF Arm
- Optimized arm's end effector to prevent **back-drive** and enable movements with millimeters-level precision using encoder feedback, **inverse kinematics**, and **PID control**, for performing tasks like autonomous keyboard typing
- Designed and implemented differential gearbox and a 4-bar linkage differential for rover stability
- Improved the rover's differential by switching from Rocker-Bogie to ORFL suspension, making it stronger
- Implemented a split-wheel design to reduce camber from nearly 7° to less than $\sim 1.5°$ reducing bending load on chassis-suspension attachment, improving its lifetime

Mechanical and R&D Lead **May 2025 - Present**

- Mentoring a team of **12 skilled individuals** working in Robotic Arm, Life-Detection and Mobility Subsystem
- Engineering a high-precision, **zero-backlash strain-wave** gearbox to replace bulkier Cycloidal Gearbox
- Implementing **explicit steering control** for four-wheel-drive to improve rover yaw motion and chassis stability
- Working on a **multilayered-Gearbox** to reduce size while increasing the reduction ratio and a **Belt-Drive System** driven by BLDC motors, to stabilize the robotic arm

Projects

Mass Function and Distance of Star Clusters using GIT Data **Aug 2025 - Nov 2025**

Prof. Varun Bhalerao | Astrophysics | Course Project | GROWTH-India Telescope Project Documents

- Procured live **GIT** observations of two open clusters and performed photometric reduction in the *u, g, r, i* bands to extract apparent magnitudes of stars and compute **stellar temperatures** from color indices (g–r, g–i, u–g)
- Estimated **cluster distances** and **absolute magnitudes** of stars by aligning the observed sequence with the standard Hertzsprung-Russell **(H-R) diagram**, and plotted absolute magnitude against temperature
- Determined the **Spectral-Type** and **Luminosity Class** of each star in order to estimate its **stellar mass**
- Constructed the **cluster mass function**; further fitted theoretical isochrones to estimate **age of the cluster**

Surface States associated with Periodic Potentials : Paper Presentation **Aug 2025 - Nov 2025**

Prof. Hridis Pal | Condensed Matter Physics | Course Project Project Documents

- Analyzed the conditions for **Shockley surface states** in finite 1-D periodic potentials, considering appropriate (decaying wavefunction) boundary conditions at crystal boundaries
- Differentiated **Shockley** and **Tamm States**, studying their origination from band inversion and edge potential
- Presented the three-dimensional extension, detailing theoretical **surface bands** and their implications for conductivity in diamond and metals

Electronic Circuit Equivalent of Mechanical Impacting System **Aug 2024 - Nov 2024**

Prof. Punit Parmananda | Non-Linear Dynamics | Course Project Project Documents

- Performed stability analysis of mechanical soft and hard impact systems and their corresponding electrical analogs
- Observed a narrow band of chaos between period 1 and 2 of displacement amplitudes, with bifurcation when distance of soft impact from natural position or system energy is changed
- Presented the work in a team of three, to a class of 65 students, and explained to them how chaos shows up

Gyroscopic Controller **Jan 2025 - Apr 2025**

Prof. Pradeep Sarin | Digital Electronics Lab | Course Project Project Documents

- Developed an angular motion-sensing controller using an **MPU6050** (accelerometer and gyroscope) and **Arduino**
- Used **Kalman Filter** to minimize the effect of sensor-noise by varying reliability of data obtained from the sensor
- Designed a 3+1 DOF maneuverable arm using servo motors and controlled it using the **self-fabricated** controller

Grover's Quantum Search Algorithm : QISKIT **Aug 2025 - Nov 2025**

Prof. Siddhartha Santra | Quantum Information and Computing | Course Project Project Documents

- Implemented **Grover's** Quantum Search Algorithm for n Qubits (2^n entries) using QISKIT library in Python
- Designed custom **oracles** and **diffusion operators** to identify various marked states in an unsorted database
- Explored practical difficulties in implementation of Grover's Algorithm in existing quantum computers and compared it to classical computers

Honors and Achievements

- Secured an All India Rank of **585** in the IIT-JEE Mains examination out of over **1.5 million** candidates [2023]
- Secured an All India Rank of **782** in the IIT-JEE Advanced examination out of **0.2 million+** candidates [2023]
- Recipient of **NTSE Fellowship**, offered to only 2000 students nationwide from the Government of India [2021]
- Currently holding **Department Rank 5** among over **60** students of Engineering Physics batch of 2027 [2025]

National Initiative on Undergraduate Sciences
NIUS is a 14-day research camp held at Homi Bhabha Centre for Science Education (HBCSE)

- Selected among **top 72** students nationwide for NIUS camp organized by TIFR
- Attended advanced sessions and lab visits on quantum optics, semiconductors, and condensed matter physics
- Selected for a year-long project on the topic "**Role of Nonlinear Resonances in Protection of Qubits**" under Prof. Sudhir Jain for exceptional performance in theoretical and practical selection examinations

Key Courses Taken

- **Physics**: Classical Mechanics, Quantum Mechanics I and II, Electromagnetic Theory, Nonlinear Dynamics, Statistical Mechanics, Waves and Oscillations, Introduction to Condensed Matter Physics, Thermodynamics, General Physics Lab, Solid State and Nuclear Physics Lab, Quantum Information Computing, Astrophysics
- **Mathematics**: Linear Algebra, Differential Equations, Complex Analysis and Integral Transform, Multivariable Calculus, Numerical Analysis
- **Electronics**: Electronics Engineering, Analog Electronics Lab, Digital Electronics and Microprocessors Lab
- **Others**: Computer Programming and Utilization, Artificial Intelligence and Data Science, Entrepreneurship and Innovation, Chemistry Lab, Management, Economics, Environmental Science

Mentoring and Organisational Experience

Secretary of Journal Club | Department of Physics | IIT Bombay **May 2025 - Present**

- Organized a series of physics seminars to expand students' engagement with diverse research domains
- Organizing a conference aimed at freshmen and sophomores to introduce them to academic research through poster presentations and speaker sessions

Teaching Experience

- *Classical and Quantum Mechanics, Physics Dept., IITB*:
  - Conducted weekly tutorial sessions, evaluated answer books and conducted additional help session before exams
- *Analog Electronics Lab, Physics Dept., IITB*:
  - Mentored weekly lab sessions, assisted students with electronics instruments and performing experiments

Technical Skills

- Software: Solidworks, Ansys, LTSpice, AutoCAD, ArduinoIDE, MATLAB, Scilab, ImageJ, SAOImage DS9
- Languages: C/C++, Python (NumPy, Pandas, Matplotlib, Scipy, Scikit), HTML, LaTeX, Markdown

Extra-Curricular Activities

- Developing an **affordable 8-dot** handheld **Braille reader** using **electromagnetic actuation**, focused on reducing manufacturing complexity to make this technology more accessible for the visually impaired [2025]
- Selected among **12 students** for 14-day long **AFS Cultural Exchange Program** out of 120+ applicants [2017]
- Completed Praveshika Purna **(Level 3)** in Synthesizer certified by **Gandharva Mahavidyalaya Mandal** [2018]
- **Vice-Chaired** the United Nations Human Rights Council (UNHRC) in CNMUN (Model United Nations) [2020]
- Ranked **3rd** in Physics Bazinga, a physics quizzing event organized by Maths and Physics club of IITB [2025]
- Achieved **2nd Position** in Institute Volleyball League of IIT Bombay among 8 competing teams [2023]

References

Prof. Kantimay Das Gupta
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Prof. Punit Parmananda
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